

Series Q Q1

Concepts: what scRNA-seq sees, how it works, how to read the plots

Using a tumour as the worked example: why we look one cell at a time, what the plots may claim, how malignancy is called, and whose number a conclusion is.

About 45 min | Prerequisites: none | Example: glioblastoma (GBM) | Deeper: Series A



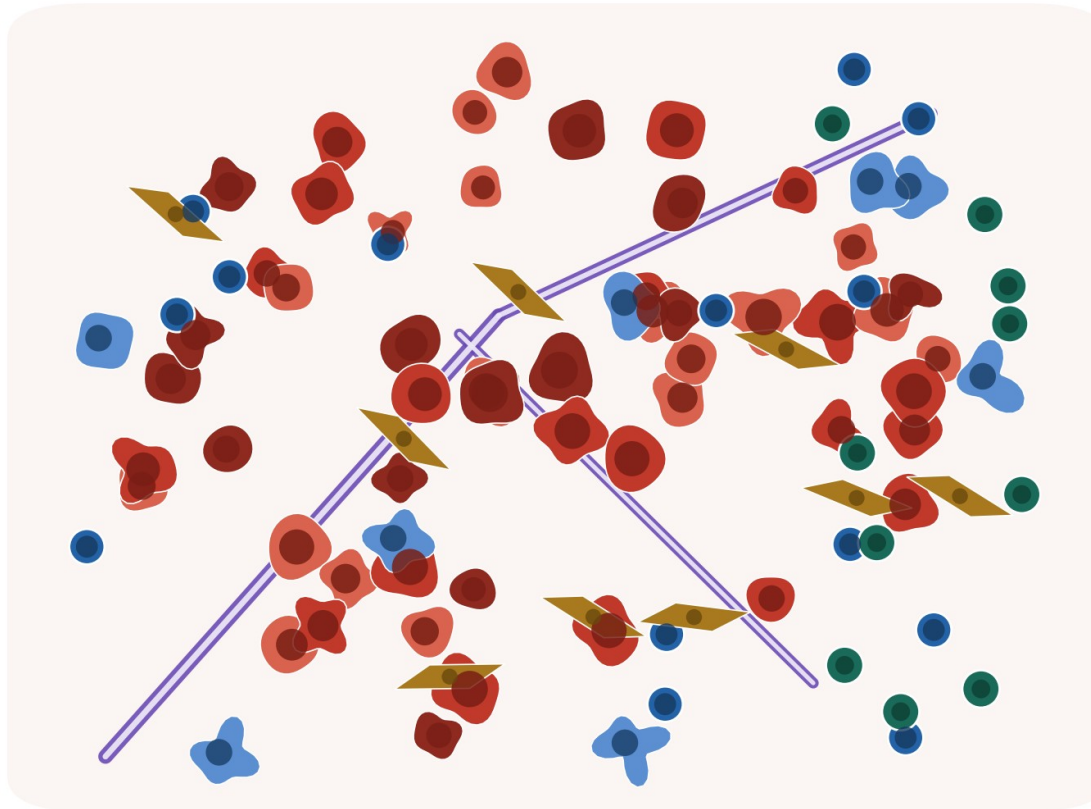
[https://github.com/Charlene717/
scRNA-seq-tutorial-Qseries](https://github.com/Charlene717/scRNA-seq-tutorial-Qseries)

A SCENE

**A 58-year-old patient, a 4 cm brain tumour on MRI.
What is actually inside the tissue the surgeon removes?**

Pathology says: glioblastoma, grade IV. That name gives us a diagnosis —
not the contents of the tissue.

A tumour is an ecosystem, not just malignant cells



one tumour section (schematic)

- malignant: several states in one tumour
- immune: macrophages, microglia, T cells
- stroma: fibroblasts, pericytes
- vascular endothelium
- invaded normal tissue (GBM: oligodendrocytes, neurons)

Proportions vary hugely by cancer type, site, even by where we sampled within the same tumour—there is no universal set of numbers. That is exactly what single cell measures and bulk cannot read

Malignant cells are only part of it, and come in several states; the rest is immune, stroma, vessels and invaded normal brain

WHERE THIS EPISODE STARTS

**Treatment failure, relapse, resistance —
the answer is usually 'which cell, in which state'.**

Answering that needs a technology that looks one cell at a time. This episode explains why, and what the data looks like.

Episode goals

1

A tumour is an ecosystem — name four questions bulk cannot answer, and say when single cell is not needed

2

Analysis lives in PC space, the UMAP is a projection — know what a UMAP may and may not claim, and read dotplots and FeaturePlots

3

How malignancy is called depends on the cancer — the preconditions of the three kinds of evidence, and when markers suffice versus when CNV is required

4

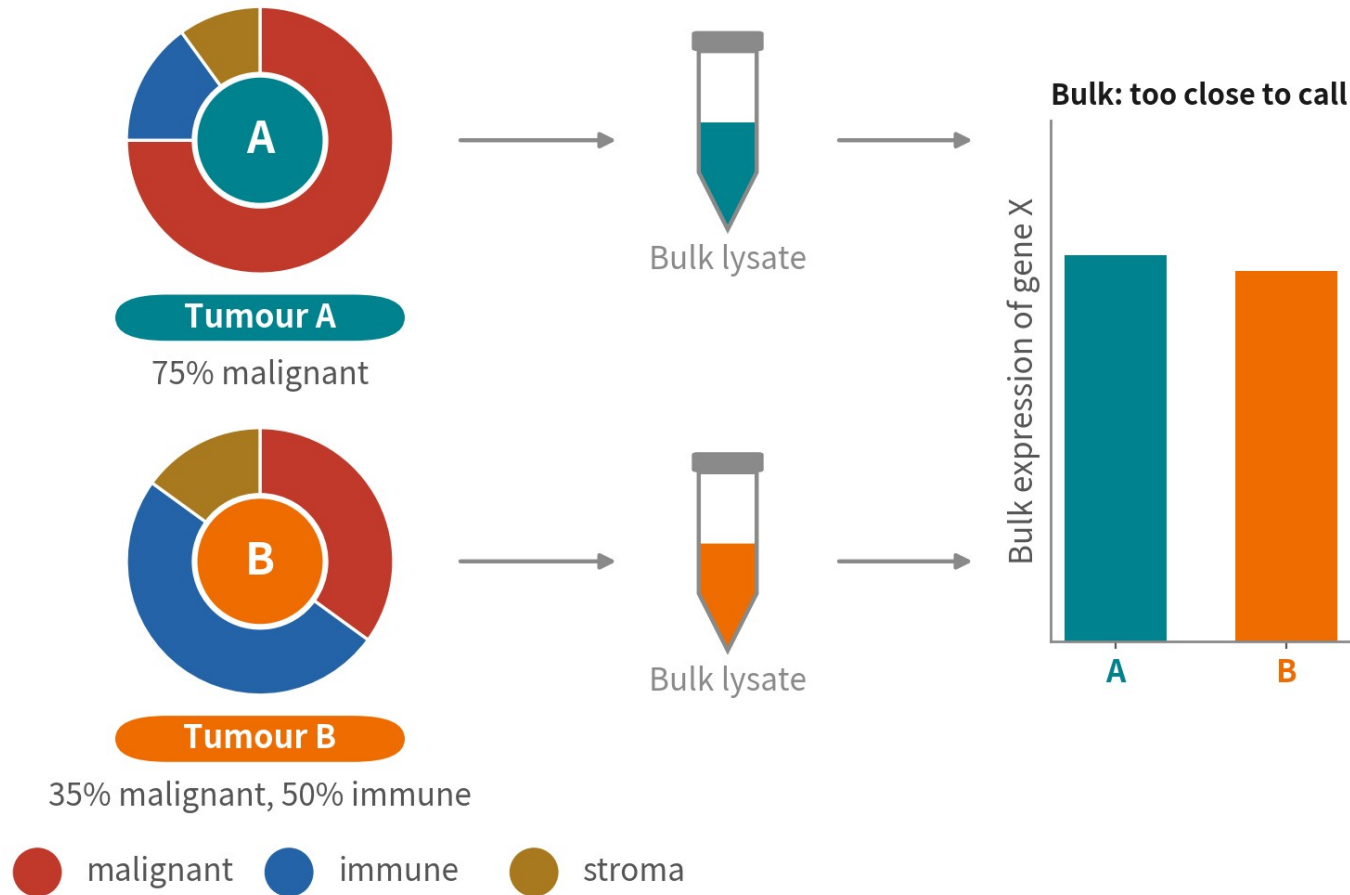
The unit is the patient — why n counts patients, not cells, and what that decides in design and in analysis

01

A tumour is an ecosystem, so we look one cell at a time

See bulk's limits first; the value of single cell — and
when it is not needed — follows

Bulk RNA-seq: grind the whole tissue, measure one average



Bulk numbers we cannot tell apart, compositions that are nothing alike

Gene X is high in malignant cells and in immune cells. Bulk sums them, so we cannot tell who produced the signal

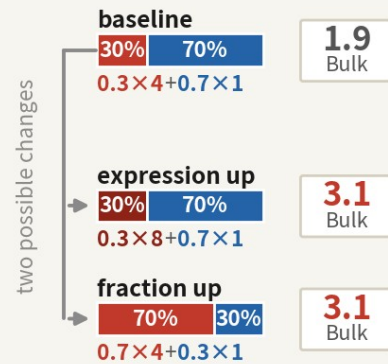
Nor in what proportion: the same bulk value can come from 75% malignant, or from 35% malignant plus half immune

In oncology this is the 'purity' problem: every bulk profile is diluted by stroma and immune cells, by an unknown factor

Two tumours with completely different composition can give bulk numbers too close to tell apart — oncology calls this the purity problem

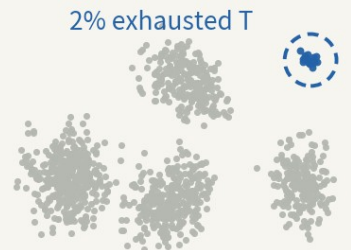
Four tumour questions bulk cannot answer

① Who produced the signal?



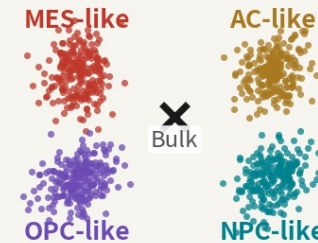
EGFR rises—did each malignant cell change, or did malignant cells get more numerous? Both land on the same bulk value

② What are the rare cells doing?



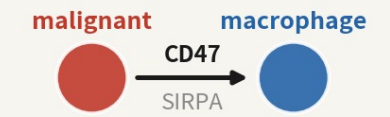
Exhausted T cells or cancer stem cells may be 1–2%. Diluted away in a bulk average; in single cell they form their own cluster

③ How many states do malignant cells have?



GBM malignant cells occupy four states at once and switch between them. Bulk gives us one average position

④ Who is talking to whom?



Bulk: both genes detected
but not who sends, who receives

Ligand and receptor are both detectable in bulk, but not who sends and who receives. Cell communication needs cell-level identity

Who produced the signal, what the rare cells are doing, how many states malignant cells occupy, who is talking to whom — all four are clinically real

When single cell is not actually needed

One test: does our question need to know who each individual cell is? If not, do not use single cell

Skip it | overall change



Overall change after treatment, prognostic subtyping, cohorts of hundreds. The answer is one tissue-level number — bulk is cheaper, steadier and better powered

Skip it | just counting



We know which types to count and have markers: flow, IHC or bulk deconvolution (Q3) all answer it, across hundreds of samples

We need it | unknown types



Rare populations with no marker, unnamed types, continuous states among malignant cells, who signals to whom. The answer IS each cell's identity — average it and it is gone

02

From a piece of tissue to a matrix

Three words are enough: GEM, Barcode, UMI

Operating table to sequencer: four steps



Dissociation

Enzymes break tissue into a single-cell suspension. Tumours are tough and necrotic; this step is brutal and leaves marks



Partitioning

A microfluidic chip seals each cell with a bead in an oil droplet (GEM)



Library

Each cell's mRNA is tagged with its bead's Barcode; everything is pooled into one library



Sequencing & counting

Cell Ranger sorts reads back to cells by Barcode, de-duplicates by UMI, then builds the matrix

More than one platform: 3', 5', full-length, nuclei

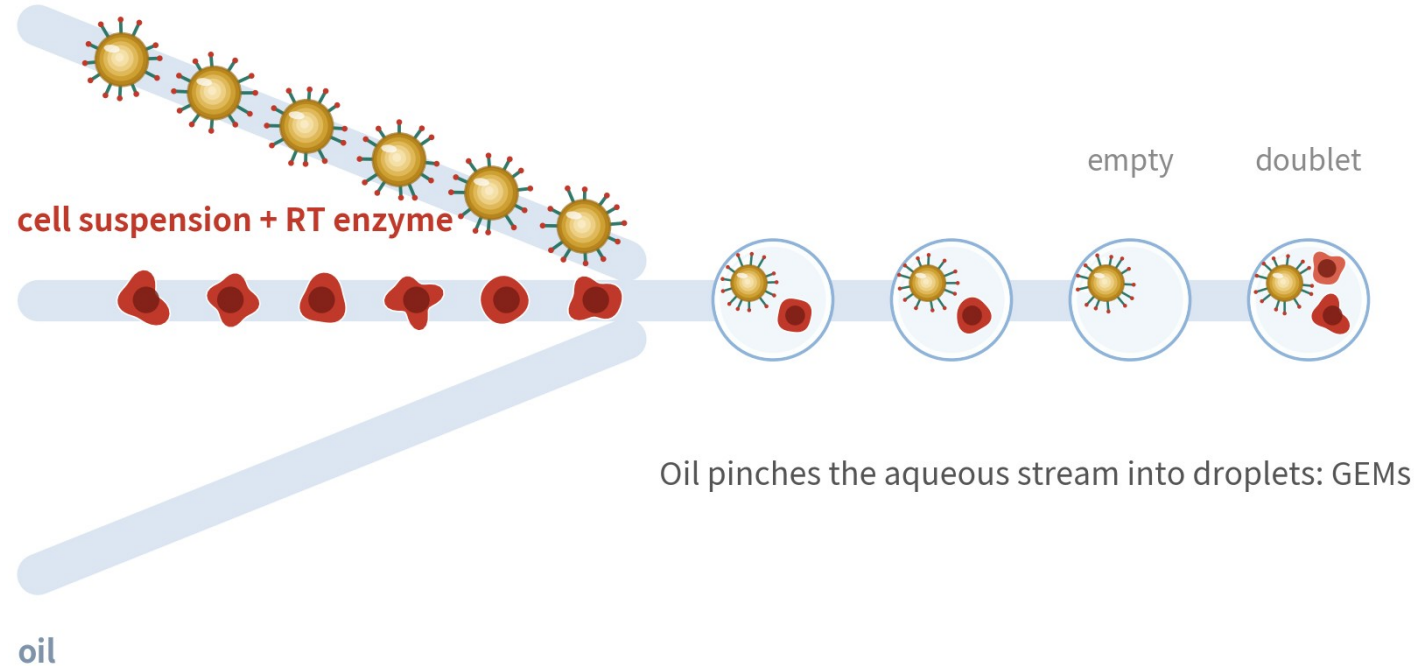
This series uses two: 10x 3' in Q2, Smart-seq2 in Q3. The differences set QC and CNV parameters

Platform	What is read	Cells / depth	Best for	Watch out
10x 3' (Q2)	3' end + UMI	5–10k cells 20–50k reads	Composition, types, states the default	Doublets, ambient RNA
10x 5' + V(D)J	5' end + TCR/BCR	as above	Immunotherapy T/B clones	No extra info on tumour cells
Smart-seq2 (Q3)	Full-length no UMI	100s–1000s ~1M reads	CNV, splicing, mutations few precious cells	Costly, slow reads ≠ molecules
snRNA-seq (nuclei)	Nuclear RNA mostly unspliced	as 10x	Frozen tissue, brain fragile tissue	mt ≈ 0 fewer immune cells

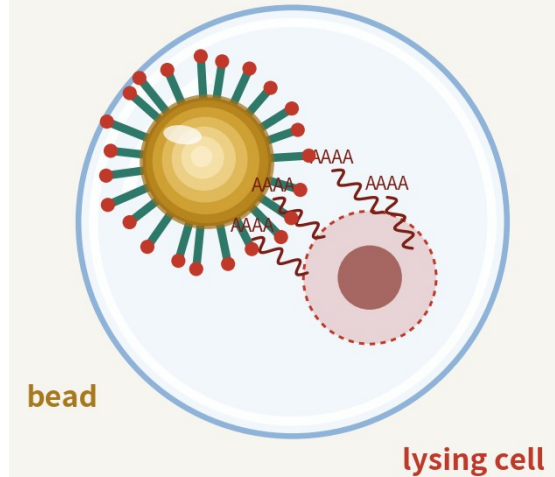
References: Picelli S, et al. Nat Protoc. 2014;9(1):171–181. doi:10.1038/nprot.2014.006 | Zheng GXY, et al. Nat Commun. 2017;8:14049. doi:10.1038/ncomms14049

A GEM: one cell, one bead, one droplet

gel beads (millions of Barcoded primers each)



Inside one GEM



**1 cell+1 bead+
lysis buffer**

The cell lyses beside the bead;
every mRNA's poly(A) tail binds
that bead's primers—its barcode
becomes the cell's ID

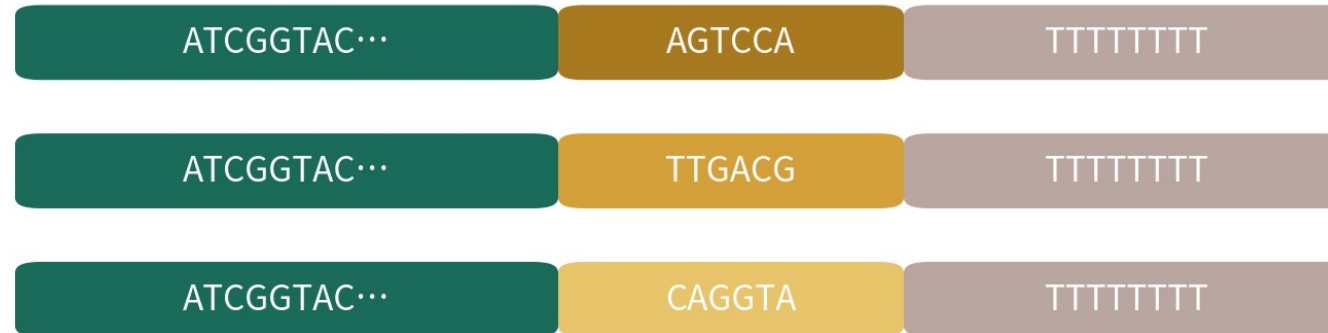
Bead-cell pairing is random: most droplets are empty, a few catch two cells. Both become QC's job later

The Barcode records which cell; the UMI records which molecule

Every primer on a bead



Same bead: identical Barcodes → which cell



Every primer a different UMI → which molecule

Why UMIs

Three identical reads:

AGTCCA

AGTCCA

AGTCCA

same UMI → PCR
copies, count 1

AGTCCA

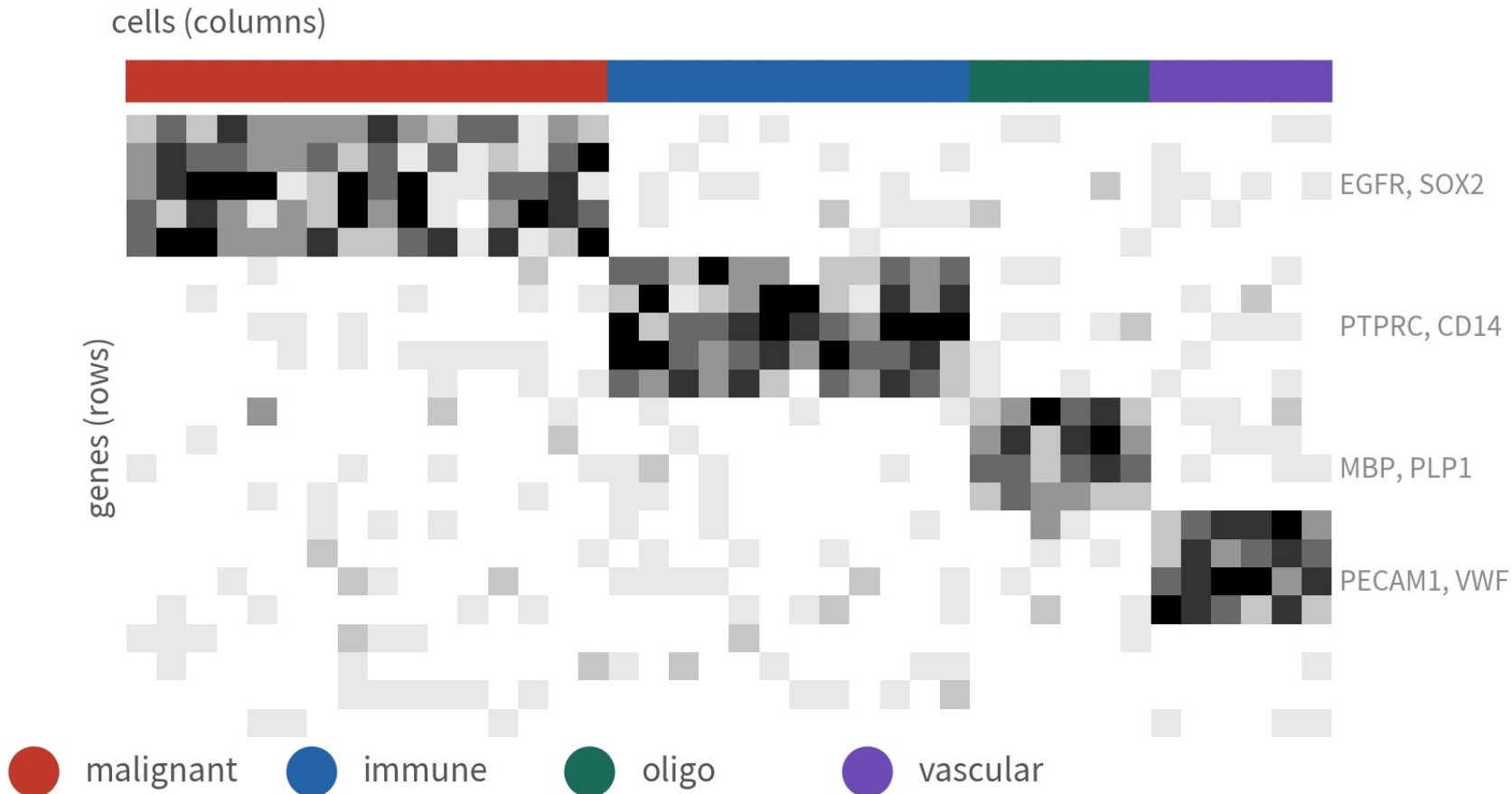
AGTCCA

TTGACG

two distinct
UMIs → count 2

The numbers in the matrix are molecules, not reads — the most fundamental difference from bulk

The output: a gene-by-cell count matrix



Three files

Barcodes .tsv

one Barcode per column

features .tsv

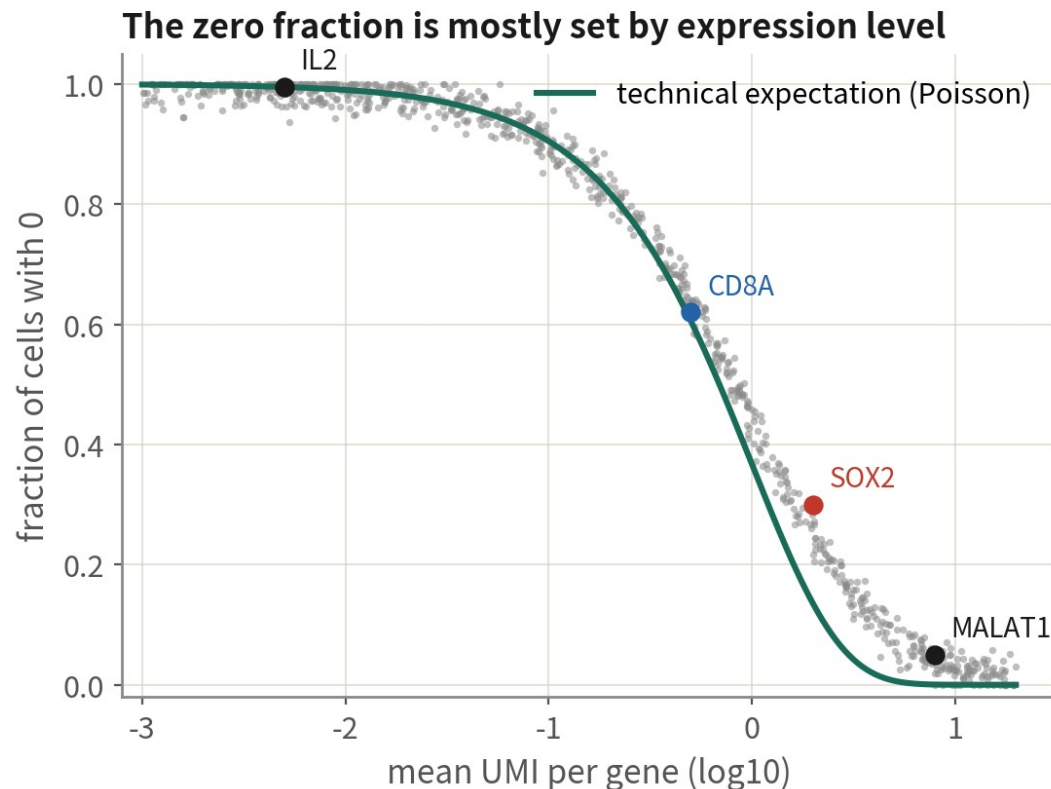
one gene per row

matrix.mtx

only non-zero
entries (sparse)

One column per cell, one row per gene; over 90% zeros — the nature of the technology, not broken data

Ninety percent zeros: two kinds of zero, and conclusions rest on groups of cells



A zero has two possible meanings

- 1 truly not expressed**
e.g. SOX2 in a T cell—this zero is signal, and it is what we want
- 2 expressed but not captured (dropout)**
Only 10–30% of a cell's mRNA is captured; low genes are often missed. This zero is noise—recovered by looking at cells as a group

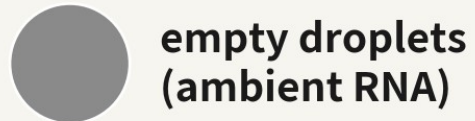
So: never conclude from one gene in one cell; conclusions rest on the distribution across a group of cells

The zero fraction is mostly set by expression level — technical expectation, not broken data. One gene in one cell means nothing; that is why every analysis looks at groups

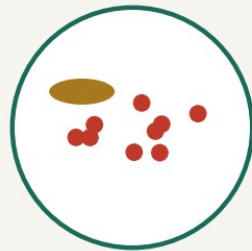
Not every column is a cell



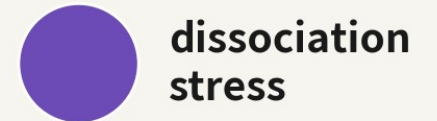
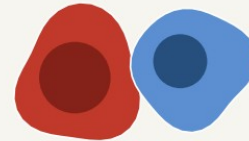
Tumour dissociation is brutal: necrotic regions, enzymatic digestion; percent.mt spikes and is often bimodal



Necrosis releases lots of free RNA; empties carry malignant genes and look like low-quality malignant cells



A malignant cell stuck to a macrophage reads as a 'tumour cell expressing immune genes'—it has happened in the literature



FOS/JUN/HSPA1A-high cells scattered across types. Not junk, but their 'activation' is an artefact



Tumour samples have plenty of all four. Each has a barcode and looks like a cell — hence QC is step one

Getting the data leaves marks: dissociation stress



What it is

A FOS/JUN/HSPA1A-high malignant subset gets written up as a 'stress-response subclone'.

Why it happens

Clean markers, significant enrichment, and it resembles a published 'MES-like state'.

What it costs you

It is the immediate-early response to enzymatic dissociation, present in every type. Tumour tissue dissociates slowly, so the artefact is strong.

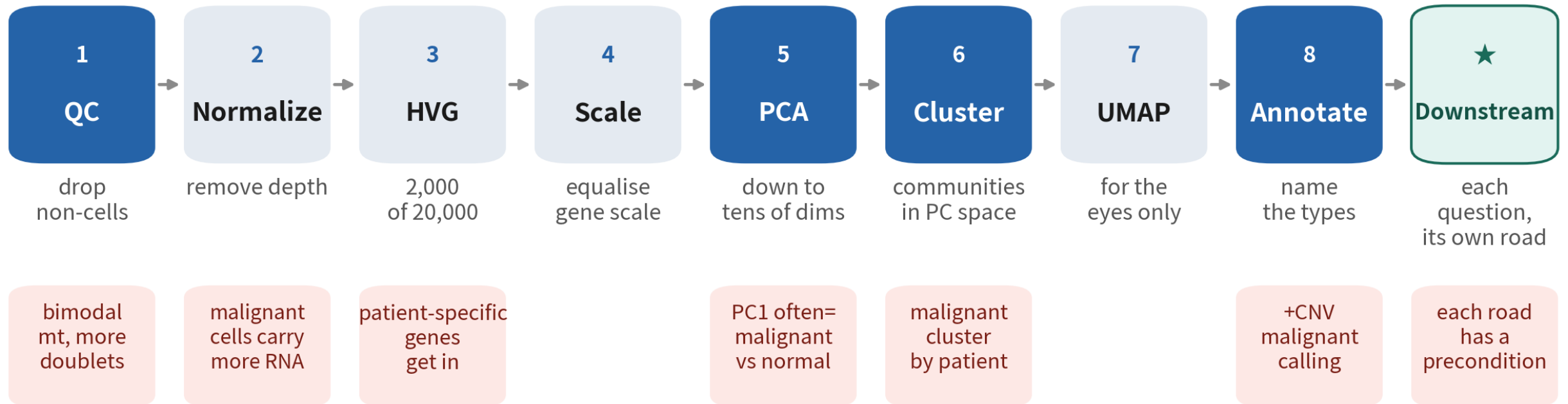
03

What the pipeline does

Do not memorise step names — remember the problem each solves; downstream analysis starts once annotation is done

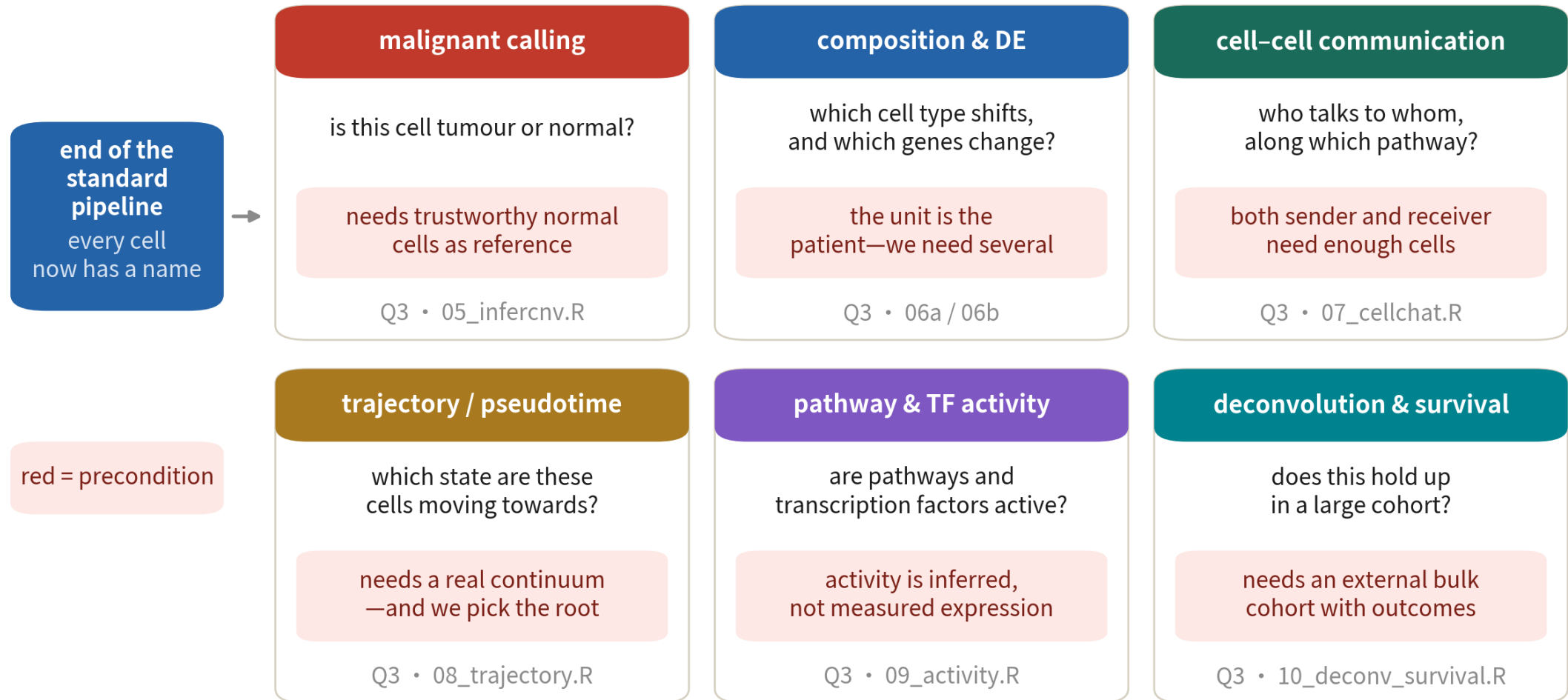
What problem each step of the pipeline solves

Dark = steps that matter most for tumour data; red cards = tumour-specific notes; green = where the pipeline ends and downstream begins



The number of steps varies with the data (SoupX, doublets, integration...) — the count is not the point. Red cards mark what is special about tumours; Q2 runs this route line by line on real GBM data

Once annotation is done: six downstream roads



The red text is each road's precondition — why 「 how many patients 」 is one of them comes in the last section

04

Analysis lives in PC space, the UMAP is only a projection

For most people this is the only skill they will ever need: knowing what a plot may and may not claim

The left side does the maths, the right side is for our eyes

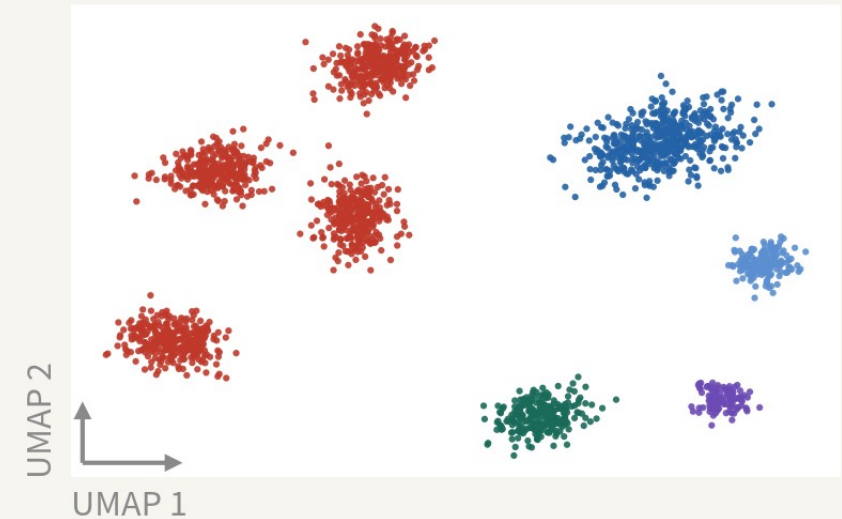
PC space (10–30 dims, invisible)

- clustering: KNN graph + communities
- cell–cell distances, neighbours
- integration (Harmony corrects here)
- trajectory, pseudotime

RunUMAP

projection

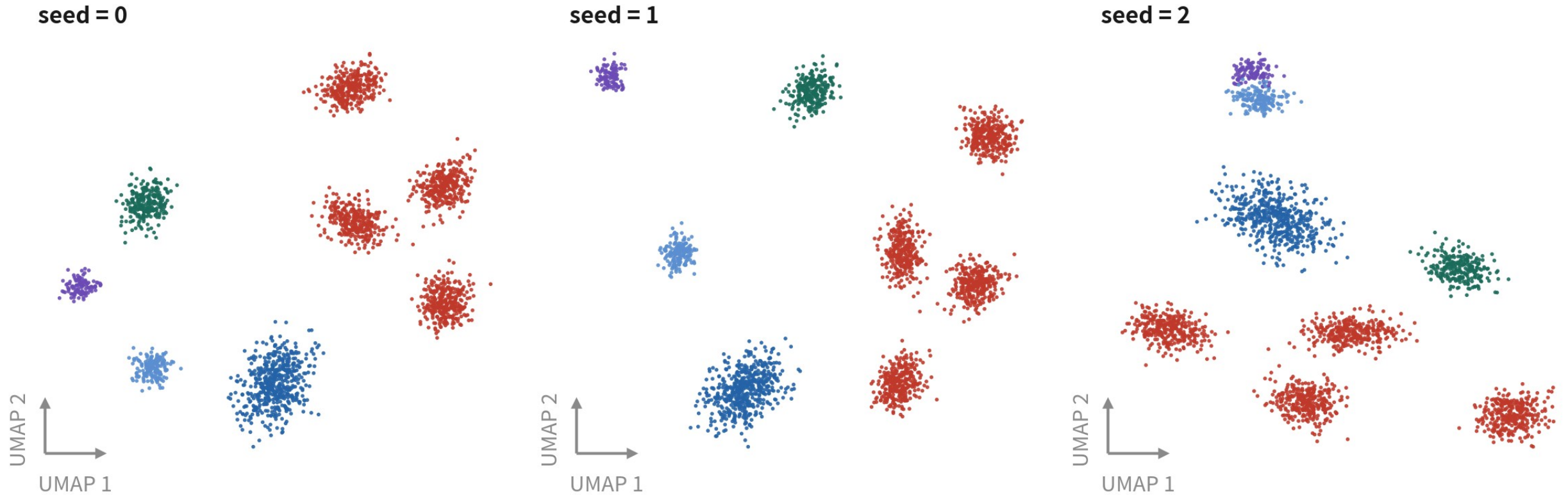
UMAP (2 dims, visible)



For looking only: which cells resemble each other — distance, area and shape are not evidence

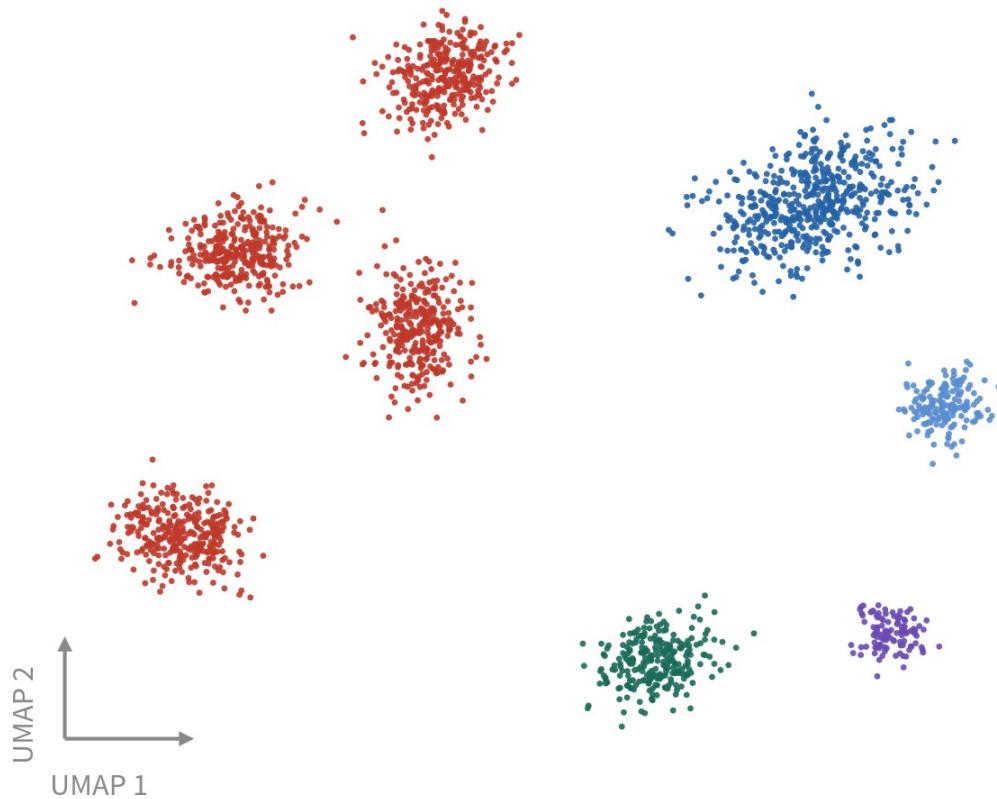
Clustering, distances, integration and trajectories all happen on the left; the right is just for our eyes

Same data, only the seed changes



Same data, same code, only the seed changes: positions and shapes all move, yet not one cell's cluster label changes — clustering finished in PC space; the UMAP is only drawn at the end

What a UMAP can and cannot tell us



It can tell us

- ✓ which cells resemble each other (local neighbours are reliable)
- ✓ roughly how many groups, which are connected
- ✓ where a gene lights up (FeaturePlot)

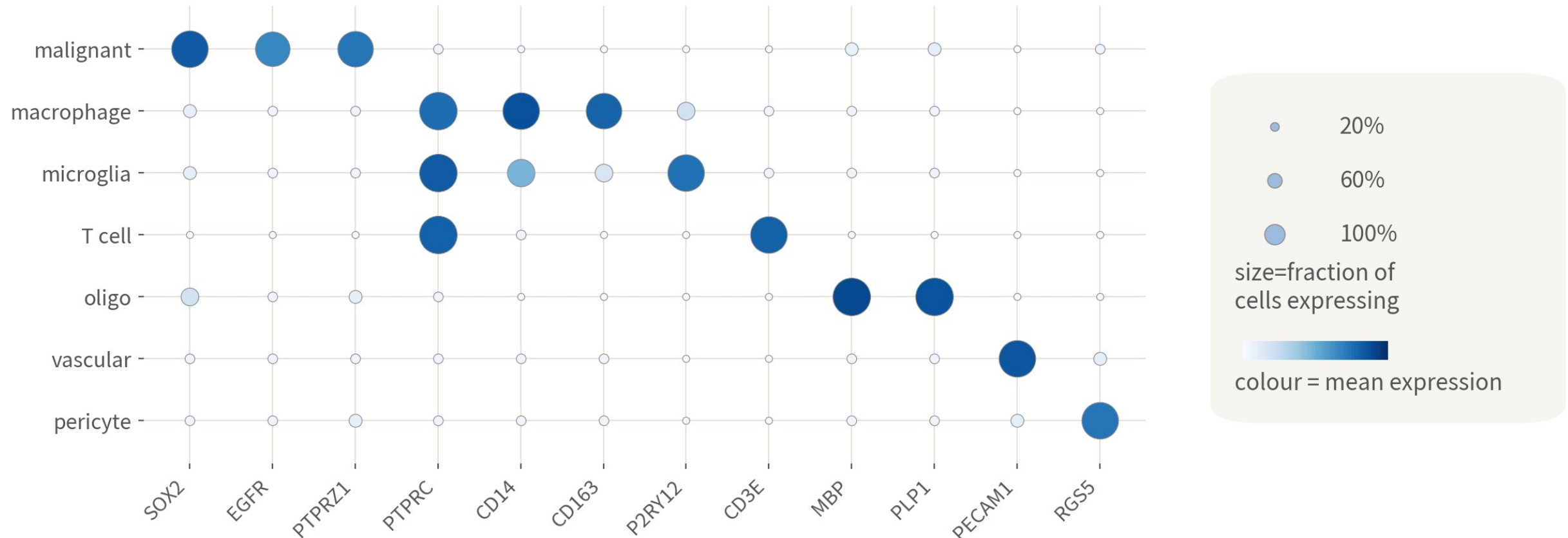
It cannot tell us

- ✗ distance $\neq$ relatedness (a new seed moves it)
- ✗ area $\neq$ cell count (use table())
- ✗ an elongated shape $\neq$ a trajectory
- ✗ four malignant clusters $\neq$ four subtypes (those are four patients)

Before reading a band as a trajectory, rule out three things: cell cycle, a sequencing-depth gradient, and a doublet bridge

Dotplot: read the two dimensions separately

Dotplot: which group expresses what — a TME marker panel



Size is the fraction of cells expressing, colour the level. Blocks along the diagonal mean a good annotation

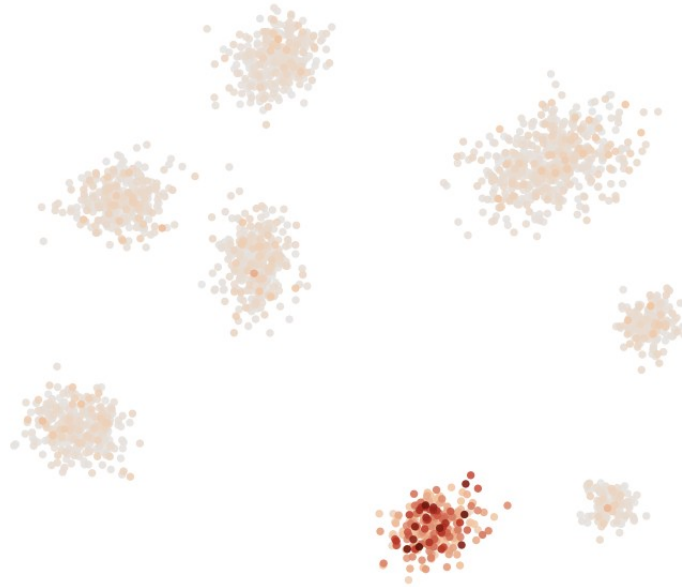
FeaturePlot: colour does not always mean expression

a clean marker: PTPRC



lights only in immune cells,
everything else grey—usable as a gate

ambient RNA: MBP



bright in oligodendrocytes, but every
cell has a little—mRNA from lysed
cells drifts into every droplet

a low-detection gene: IL2



a scatter of single bright
cells, no cluster shape—no
conclusion from single cells

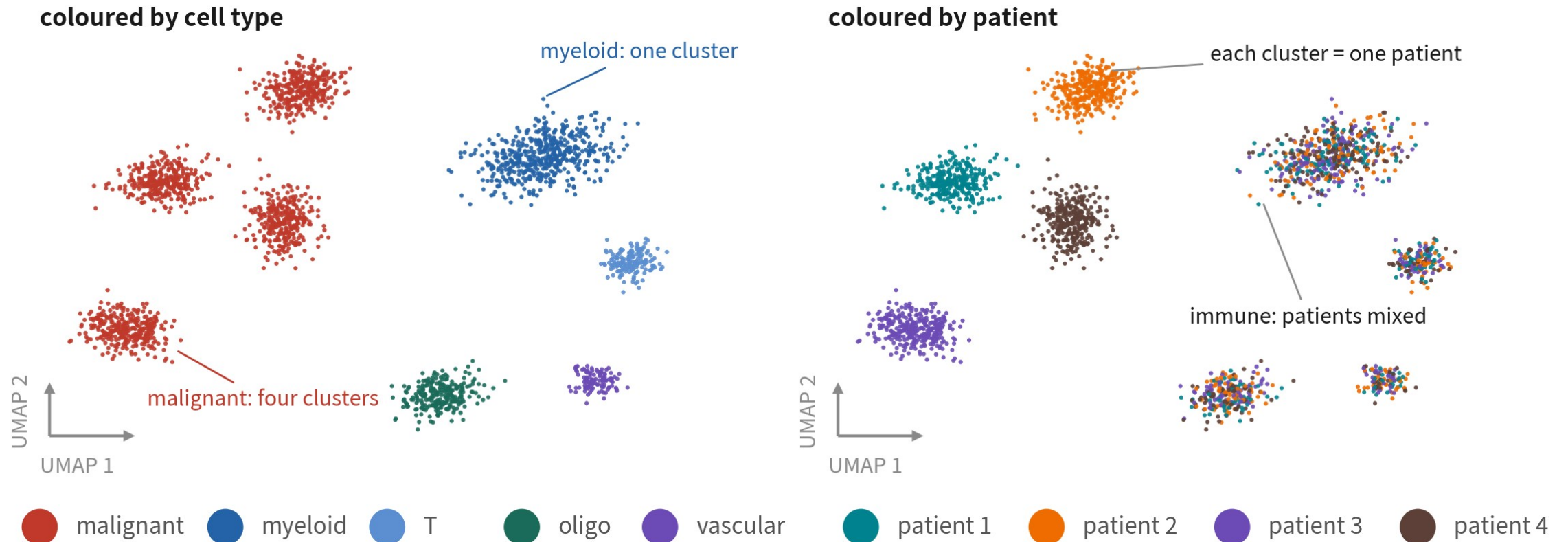
A clean gate lights one block; ambient RNA gives every cell a little; a low-detection gene is a sprinkle. First question: does the bright region have the shape of a cluster?

05

How malignancy is called depends on the cancer

Ask one thing first: is the normal counterpart of this tumour mixed into the sample

Why calling it is hard: malignant cells cluster by patient, normal cells mix



Each patient's tumour carries its own CNV and mutation set, so its malignant cells usually form their own cluster. This is more likely biology than batch; check with normal cells as a control before integrating (Q3)

Three kinds of evidence for malignancy, each with its own precondition

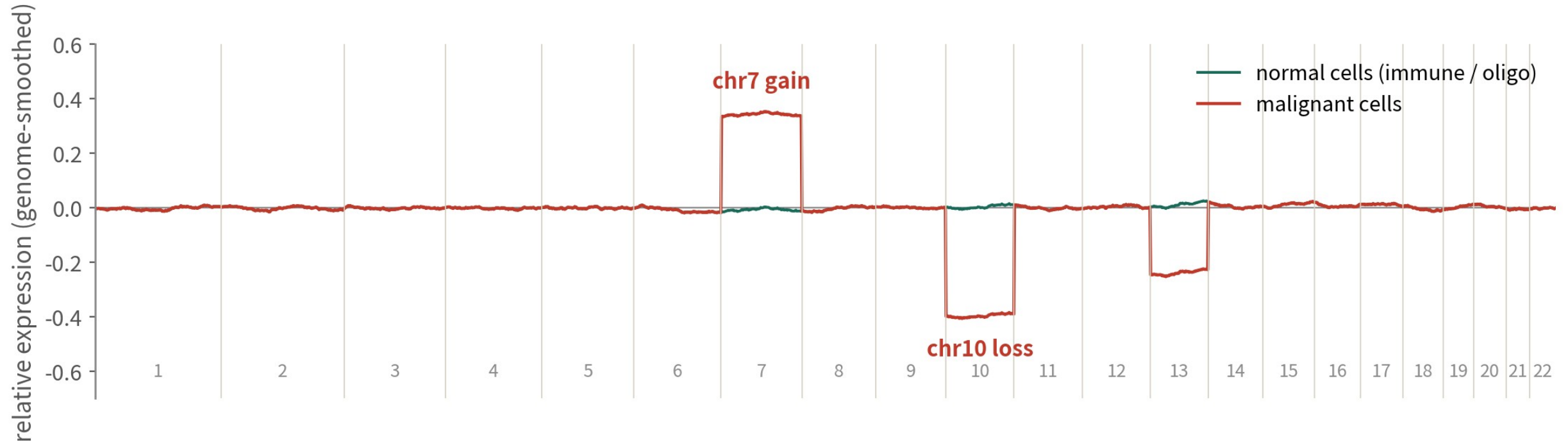
There is no universal answer. What decides is whether the normal counterpart is in the sample, and what your sequencing can read

Evidence	Strength and limits	When it suffices
Defining variants IDH1 R132H H3K27M, EGFRvIII fusions	Strongest — diagnostic. But these are point mutations or specific variants, and 10x 3' reads only the transcript end, so it usually misses them	Use it directly when full-length (Smart-seq2) or targeted data exists
Lineage markers EPCAM/KRT, SOX2/EGFR...	Gives lineage only. Whether that equals malignancy depends on whether same-lineage normal cells are present	Often enough in epithelial tumours (lung, colorectal, breast, ovary), where normal epithelium is scarce; still worth confirming with CNV
Inferred CNV inferCNV, CopyKAT	Genome-level evidence. Needs a trustworthy normal reference; fails in low-CNV tumours (IDH-mutant low-grade glioma, some paediatric tumours)	In tumours like glioma, where the normal counterpart sits right next door, this is usually what separates them

References: Patel AP, et al. Science. 2014;344(6190):1396–1401. doi:10.1126/science.1254257 | Gao R, et al. Nat Biotechnol. 2021;39(5):599–608. doi:10.1038/s41587-020-00795-2

CNV: a whole chromosome arm shifting together

A malignant cell's ID is its genome: infer CNV from chromosome-scale expression trends



One gene says nothing, but a hundred neighbouring genes on the same chromosome moving together has one explanation: the copy number of that DNA changed. This is how inferCNV works — and the only reliable basis in tumours like glioma, where the normal counterpart cell sits right next door

The four major groups: three are easy, one is not

The first cut in annotating tumour data is always these four

Three: immune · vascular · oligo

Naming ends here

- Immune — PTPRC
- Vascular — PECAM1, VWF
- Oligo — MBP, PLP1 (the inferCNV reference)

The fourth: glial / malignant

Naming is not the end

- Glial — SOX2, OLIG2, PTPRZ1, EGFR
- Malignancy needs CNV + patient specificity
- Call first, then states (MES/AC/OPC/NPC)

06

The unit is the patient, not the cell

The first three decide what we see; this one decides what we may claim

Thirty thousand cells from one patient is still $n = 1$

Both designs total 30,000 cells; only the number of patients differs

Design 1: one patient, 30,000 cells



$n = 1$

statistical unit

Design 2: six patients, 5,000 each



$n = 6$

supports patient-level inference

30,000 cells from one patient is still $n = 1$. Testing cells as samples inflates the degrees of freedom into false significance — Q3 measures how big the effect is

The most common way to cross it: counting cells as samples

What it is

One patient's tumour vs one patient's normal brain,
a test over 30,000 cells, 3,000 'disease genes'.



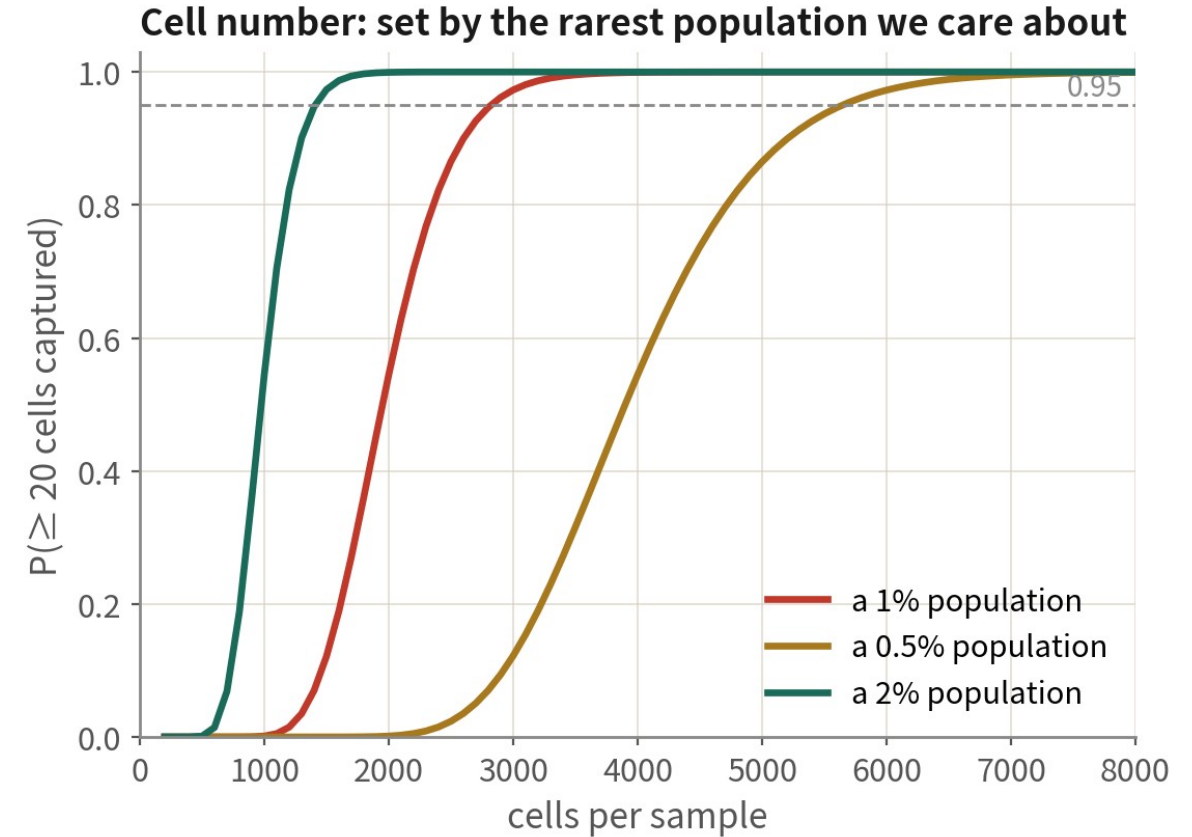
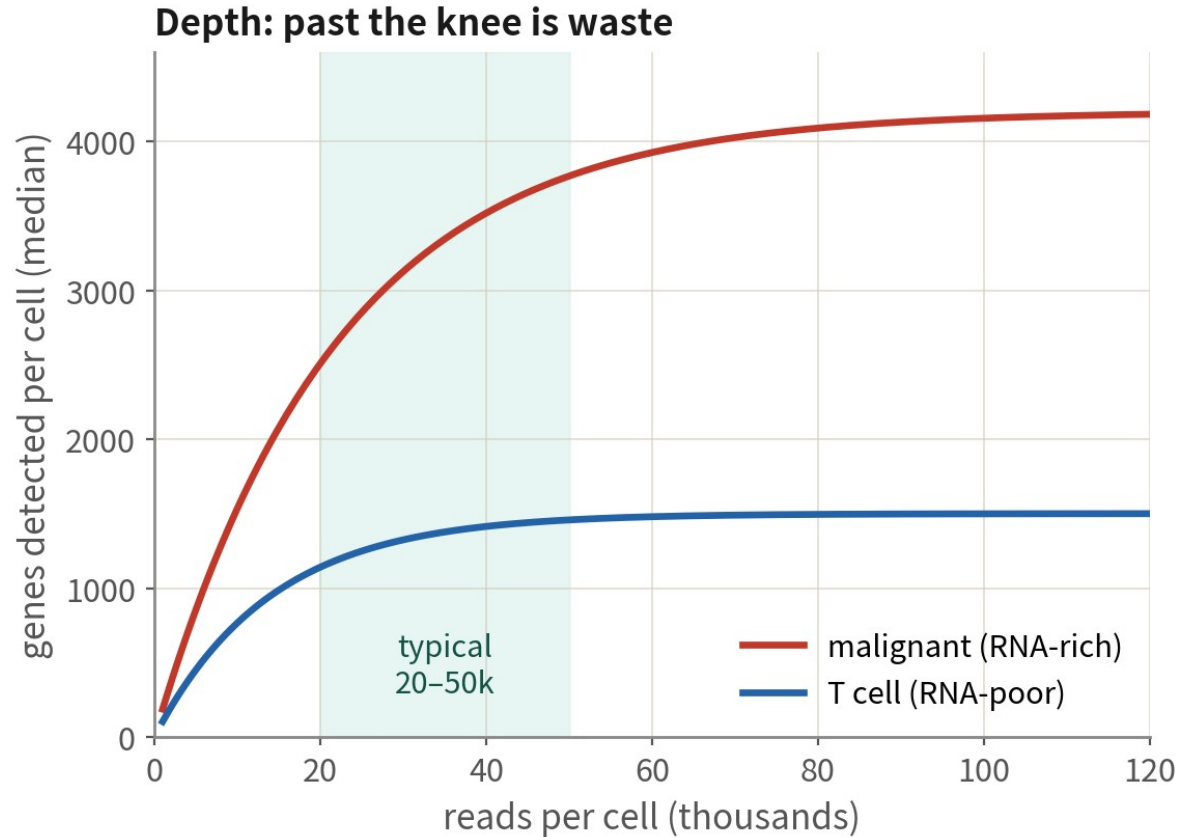
Why it happens

Every cell looks like an independent observation.

What it costs you

Cells from one patient are correlated; n is really 1.
Most of those genes will not hold in the next patient.

Three numbers in study design: patients, cells, depth



Decide in this order: patients (power), then cells (rarest population), then depth (waste past the knee). On a fixed budget, one more patient nearly always beats 10,000 more cells

Reading Figure 1 of a single-cell paper: five questions first

All five answers live in the methods and supplements; if we cannot find one, discount the conclusion

Question	Why it matters	Where to look
How many patients per group?	n is patients; between-group claims at $n < 3$ are descriptive	Table 1, 'Samples' in methods
How were cells QC'd and doublets removed?	Without doublet removal, 'dual-lineage' types are usually artefacts	'QC' in methods, Fig S1
What was integrated, what was corrected away?	Integrating by patient flattens real malignant differences (Q3)	'Integration' in methods, pre/post UMAPs
How was malignancy called?	'Tumour cells' by markers alone are not credible; look for CNV or mutations	inferCNV / copyKAT heatmap
Is DE done per cell or per patient?	Cell-level tests hand us thousands of false positives (Q3)	'Differential expression': pseudobulk or not

What this technology can and cannot answer

What it cannot do is not a flaw but the edge of what we may claim — better written by us than by a reviewer

CAN ANSWER

What single cell gives

- Which cell type the signal came from — composition change vs per-cell change, which bulk cannot separate
- What rare populations are doing — exhausted T cells, stem-like cells, diluted away by averaging
- How many states one cell type holds — the four GBM malignant states; bulk gives only the average
- Who may be talking to whom — ligand-receptor inference (inference, not proof of contact)
- No need to assume which populations exist — they emerge from the data
- One experiment feeds everything downstream — malignancy, trajectories, communication, deconvolution

CANNOT ANSWER

The edge of what we may claim

- Position — lost at dissociation; asking about it needs spatial transcriptomics
- Anything but RNA — protein, modifications, metabolites are absent; high mRNA is not high protein
- The in vivo state — what we measure is the cell after half an hour in enzyme
- Point mutations, splicing, fusions — 10x 3' reads only ~90 bases from the end; hence CNV as the fallback
- Time — each cell is measured once; 'A becomes B' is inference, and a trajectory is an ordering, not time
- Power — more cells never buys n; statistical power comes from the number of patients

Four rules to keep before touching the data

The first line is the principle; below it is what the principle looks like in practice

1. A tumour is an ecosystem



Ask who produced the signal before asking how much it differs

When the question does not need each cell's identity, bulk is cheaper and better powered

3. Malignancy calling depends on the cancer



Normal counterpart in the sample → CNV is required; absent → markers can do

Triangulate whenever possible; never rest on one kind of evidence

2. The UMAP is only a projection



Clustering, distances, integration and trajectories never use its coordinates

Count area with `table()`; for a band, first rule out cycle, depth and doublets

4. The unit is the patient



In design it sets how many patients; in analysis it sets the level of the test

Check each downstream road's precondition; when it fails, say so instead of drawing the plot

What lies beyond single-cell transcriptomes

This series reads RNA only; knowing what else exists tells us what we are missing

Multimodal



RNA + protein / chromatin. CITE-seq adds surface proteins (sharper immune calls); 10x Multiome adds ATAC (regulation, lineage). Same analysis skeleton as this course

Spatial transcriptomics



Visium, Xenium, MERFISH keep position: who sits by vessels, in necrosis, at the core-edge boundary. Single-cell data often serves as the reference for deconvolution

Genome and lineage



TCR / BCR sequencing tracks clones; scDNA-seq or mitochondrial variants track subclones. Tumour-evolution studies pair these with expression

Check yourself 1

Bulk shows CD8A three-fold higher in tumour A than B.
The most conservative reading?

- A A's T cells are more cytotoxic
- B A may have more CD8 T cells, or each expresses more — Bulk cannot tell
- C A will respond better to immunotherapy

Answer B. Composition change and intrinsic change look identical in bulk. A and C over-interpret; we need single-cell data (and multiple patients) to move in that direction.

Check yourself 2

Four GBM patients: malignant cells form four clusters, macrophages one blended cluster. This means?

- A Severe batch effect; integrate
- B Four malignant subtypes
- C Expected: malignant cells carry patient-specific CNV, normal cells do not

Answer C. Macrophages blend well, so technical batch is small; the malignant clusters come from genomic differences. Forcing integration erases real patient biology.

Check yourself 3

The strongest evidence that a cluster is malignant?

- A High EGFR and SOX2
- B Whole chromosome arms (e.g. chr7 gain, chr10 loss) shifting together
- C Sitting far from immune cells on the UMAP

Answer B. Markers only give lineage and UMAP distance is not evidence; CNV is genome-level. Note this asks about GBM — in epithelial tumours the normal counterpart is scarce and markers often suffice (see question 6).

Check yourself 4

After clustering, a FOS/JUN/HSPA1A-high group appears — with a patch inside malignant, macrophage and oligodendrocyte cells alike.

Most likely?

- A A cross-lineage 'stress-response subpopulation'
- B The immediate-early response to dissociation — a technical artefact
- C Low-quality cells that QC should have removed

Answer B. The test is that it appears across types: a real biological subpopulation usually belongs to one lineage, while a signal present in every type is almost always a handling artefact.

Check yourself 5

Fixed budget, one choice: one more patient, or raising each sample from 5,000 to 15,000 cells. The question is core vs periphery differences in malignant cells — which?

- A One more patient
- B More cells per sample
- C Equivalent — take whichever is easier

Answer A. The unit here is the patient and power comes from patient count; more cells only measure the same patient more finely. More cells pay off only when hunting a rare population.

Check yourself 6

In which situation do lineage markers alone suffice to call malignancy?

- A Any solid tumour, as long as the markers are clean
- B When same-lineage normal cells are essentially absent — e.g. epithelium in lung or colorectal cancer
- C Never — a CNV heatmap makes markers unnecessary

Answer B. Markers give lineage, not malignancy. In epithelial tumours normal epithelium is scarce, so EPCAM+ is nearly synonymous with tumour; glioma is the opposite — reactive astrocytes and OPCs sit right next door, so CNV is required.

NEXT

Q2: the single-sample standard pipeline — from raw matrix to cell annotation

Concepts done — now the hands-on. Next episode runs 10x's public GBM 5k (5,604 cells) from loading to annotation, with three practice scripts — every block copy-ready.

